

PBSMS

Tenant AI Assistant

SRS v2.1 Addendum · Chapter 47 · Version 2.0 · Model A of two AI specifications

Item	Value
Document	PBSMS Tenant AI Assistant — SRS v2.1, Chapter 47
Version	2.0 — Adopted Baseline
Status	ADOPTED BASELINE — APPROVED FOR BUILD. Adopted 27 August 2026. Build authorisation is staged; see §47.0.2.
Requirement prefixes	FR-AIT- (functional) · NFR-AIT- (non-functional) · TEN-05x (tenancy) · DP-1xx (data protection)
Governs	A per-tenant, subscription-gated AI assistant operating solely on the subscribing tenant's own data
Depends on	Ch 13 (access scope) · Ch 23 (finance model) · Ch 34 (frontend) · Ch 40 (retention) · FR-ANL-040 (AI-drafts / human-approves precedent)
Companion	Model B — Internal Engineering Agent, v2.0. Separate document. Not part of the product and not an SRS chapter.

47.0 Approval and Change Control

This chapter is adopted into the SRS v2.1 baseline as of 27 August 2026 and is approved for build. It carries the same standing as any other chapter of the SRS: it is binding on implementation, and where an implementation and this chapter disagree, this chapter wins until it is formally amended.

47.0.1 Amendment process

- Any change to a requirement in this chapter requires a version increment and re-adoption. Requirement IDs are permanent — a withdrawn requirement is marked withdrawn and retained, never deleted and never reused.
- New requirements continue the existing numbering. Gaps in the sequence are expected and are not errors.
- Any implementation that departs from the literal text here shall be recorded as a named deviation with its reasoning, on the pattern established in the Frontend Design Specification §14, and approved by whoever holds the Engineering Lead role before it ships. A silent deviation is treated as a defect regardless of whether the resulting behaviour is better.
- The blocking thresholds in §47.15.2 are not amendable downward. They may be raised. Lowering any of them requires re-adoption of this chapter as a whole, not a local amendment, because their whole function is to be inconvenient at the moment someone wants to move them.

47.0.2 Build authorisation

Approval authorises the build. It does not clear every stage at once, because three prerequisites are external to engineering and cannot be closed by starting work. Stages are listed in §47.17; their authorisation status is below.

Stage	Scope	Authorisation
1	Retrieval layer under RLS, scope enforcement, audit logging — no model	CLEARED — begin immediately. No external dependency.
2	Evaluation harness, golden sets, adversarial corpus, CI gates	CLEARED — begin immediately, in parallel with stage 1.
3	Ask capability, staff roles, internal reference tenant	GATED on a model provider contractually satisfying DP-103 (zero retention, no training, no human review) and DP-107 (disclosed processing jurisdiction).
4	Shadow period, then canary tenants	GATED on stage 3 and on §47.15.2 thresholds green.
5–7	Draft, then demo mode and metering, then Explain and Find	GATED on the preceding stage. Pricing decisions in §47.11 required before stage 6.
8	Parent View exposure	NOT AUTHORISED by this adoption. Requires a separate decision and a separate risk review, as stated in §47.12 and §47.17.
—	FR-AIT-060 multilingual capability	NOT AUTHORISED until an evaluation set exists for Twi, Ga, Ewe and Hausa in this domain. Specified, not cleared.

Stages 1 and 2 are cleared deliberately and should start first. They are the isolation boundary and the evaluation harness — the two things that cannot be retrofitted — and neither depends on the provider decision that gates everything after them. Building them while procurement runs is the correct use of the interval, not a workaround for it.

47.0.3 What this approval does not settle

The open questions in §47.18 survive adoption. They are recorded there rather than resolved here, and §47.18 remains the live list. Adoption means the requirements are binding, not that every unknown has been closed — a baseline that waited for that would never be adopted.

47.1 Purpose and Position in the System

The Tenant AI Assistant answers questions and drafts text using only the subscribing tenant's own records, under the requesting user's own permissions. It is an assistive layer sitting above the existing modules. It is not a system of record, not an authority, and not a decision-maker.

Two commitments already in the SRS govern everything in this chapter:

- FR-ANL-040 already establishes the pattern — AI drafts, a human approves before anything is issued. This chapter extends that precedent rather than inventing a new trust model.
- Chapter 13 defines access scope from tenant down to record ownership. The Assistant inherits that scope wholesale. It can never see more than the person asking it.

Anything this chapter does not explicitly permit, the Assistant does not do. The prohibitions in §47.4 are requirements of equal standing to the capabilities in §47.3, and are more important to implement correctly.

47.1.1 Why the negative requirements come first in review

This product holds the personal data of minors, the financial records of fee-paying households, and academic results that determine BECE candidacy and CSSPS placement. An assistant that is merely useful is not a good outcome here; an assistant that is useful and cannot be made to say something false about a child's money, attendance or results is the only acceptable outcome. Every capability below is therefore specified with its failure mode named, and the evaluation gates in §47.15 treat groundedness, isolation and numeric fidelity as pass/fail release blockers rather than quality metrics to improve over time.

47.2 Scope

In scope	Out of scope
Natural-language questions over the tenant's own records	Any access to another tenant's data, in any form, including aggregated or anonymised
Drafting guardian messages, report-card comments, term summaries and follow-up notes for human approval	Sending, publishing or finalising anything without a named human approver
Explaining how a displayed figure was derived, by walking the records behind it	Computing, deriving or restating any monetary or grade figure
Natural-language search across records within the user's permission scope	Any write path to any tenant record
A capped, synthetic-data demonstration mode for prospects	Predictions, risk scores or characterisations of individual students
Per-tenant enable/disable and per-role restriction	Modification of its own scope, prompts, permissions or configuration

47.3 Capabilities

Four capabilities. Each is defined by what it returns, not by what it feels like to use.

47.3.1 Ask

- **FR-AIT-010** — The Assistant shall answer natural-language questions over the requesting tenant's records, restricted to the requesting user's permission scope as defined in Chapter 13.
- **FR-AIT-011** — Every factual claim in a response shall carry a reference to the record or records it was derived from, and the interface shall render those references as navigable links. A claim that cannot be traced to a retrieved record shall not be returned.
- **FR-AIT-012** — Responses returning a set of records shall state the size of the set and whether it was truncated. A response that silently omits matching records is a defect.

Worked example — a headmaster asks which JHS 2 students are below 80% attendance this term:

You asked: JHS 2 students below 80% attendance, Term 2

4 students, of 61 enrolled in JHS 2. Retrieved 11:04, today.

Ama Mensah	JHS 2A	71%	32 of 45 days	→ open
Kofi Boateng	JHS 2A	76%	34 of 45 days	→ open
Yaa Owusu	JHS 2B	64%	29 of 45 days	→ open
Kwesi Antwi	JHS 2B	78%	35 of 45 days	→ open

Source: attendance register, Term 2, 2026/2027 · 45 marked days
 [Open absence follow-up queue]

Note what the example does not do: it does not say who is "at risk", does not rank by concern, and does not suggest a cause. Those are judgements about children (FR-AIT-104).

47.3.2 Draft

- **FR-AIT-020** — The Assistant shall draft text for human review in these places only: guardian messages (Ch 26), report-card comments (FR-DOC-010), term summaries, absence follow-up notes (FR-ATT-040), and admission decision letters (FR-ADM-030).
- **FR-AIT-021** — Every draft shall enter the existing approval control for that artefact. The Assistant shall not send, publish, or save-as-final any content.
- **FR-AIT-022** — Drafts shall be marked as AI-drafted in the approval interface and in the audit record, and shall remain visually distinguishable from human-authored content until approved.

47.3.3 Explain

- **FR-AIT-030** — The Assistant shall explain how a displayed figure was derived by walking the records that produced it — a fee balance to its invoices, allocations and payments; a term aggregate to its component assessments and weights.
- **FR-AIT-031** — The Assistant shall read the value computed by the owning module and render it unchanged. It shall not recompute, re-derive or reconcile any figure (see FR-AIT-101).

47.3.4 Find

- **FR-AIT-040** — Natural-language search across students, guardians, staff, invoices, payments and messages within the user's scope, returning records rather than prose.

47.3.5 Session and context

- **FR-AIT-050** — Conversation context shall be retained within a single session, scoped to one user, one tenant, one school, one campus and one academic year.
- **FR-AIT-051** — Changing any of the four context values in the Staff Console context switcher (Frontend Design Specification §6.1) shall terminate the session and begin a new one. Context bleeding across a school or year switch would produce answers about the wrong school, which is the exact failure the four-part switcher exists to prevent.
- **FR-AIT-052** — Sessions shall expire with the user's authenticated session and shall not be resumable across logins.

47.3.6 Language

- **FR-AIT-060** — English by default. Where a tenant enables it, the Assistant shall accept questions in Twi, Ga, Ewe or Hausa and answer in the language of the question.
- **FR-AIT-061** — Parent-facing text drafted in a language the approving staff member has not confirmed they read shall carry a warning at the approval step. An approver cannot take accountability for text they cannot read.

FR-AIT-060 shall not ship on assertion. No evaluation set exists for these languages in this domain, and translation quality in Ghanaian languages for school-administrative register is unverified. It is gated on §47.15 evaluation the same as every other capability; see §47.18.

47.4 Prohibited Behaviours

These are requirements, not guidance. Each is testable and each is a blocking gate in §47.15.

- **FR-AIT-100** — No writes. The Assistant shall have no write path to any tenant record. It cannot mark attendance, enter or alter scores, publish or reopen results, create or void invoices, capture, allocate or reverse payments, alter permissions, or change configuration. Its output is text and record references only.
- **FR-AIT-101** — Not a second source of financial truth. The Assistant shall not compute, derive, estimate or restate any monetary figure. Every amount it displays shall be read from the finance module's computed value and rendered unchanged. Chapter 23.1 makes the transaction ledger the single source of every balance; an AI-produced figure disagreeing with the ledger is a severity-one defect, not a rounding difference.
- **FR-AIT-102** — No unpublished results. Draft, in-review and returned results (FR-RES-010) shall be excluded from Assistant retrieval for parent and student users unconditionally. For staff users, any draft result appearing in a response shall be labelled as draft. FR-RES-020's publication gate is not bypassable by asking a question.
- **FR-AIT-103** — No determinations. The Assistant shall not generate a grade, a promotion recommendation, an admission outcome, a disciplinary finding, or any other determination affecting a child's academic record or standing. It may summarise such a determination after a human has made it.
- **FR-AIT-104** — No profiling of children. The Assistant shall not generate predictions, risk scores, behavioural characterisations or comparative judgements about an individual student. "Which students are below 80% attendance" is a query over recorded facts and is permitted. "Which students are likely to fail" is a prediction about a child and shall be refused with a stated reason.
- **FR-AIT-105** — No cross-tenant anything. No answer, aggregate, benchmark, comparison or example shall incorporate data from any tenant other than the requesting one, including in anonymised or aggregated form. Where a tenant asks how they compare to other schools, the Assistant shall state that the platform does not provide cross-school comparison.
- **FR-AIT-106** — No self-modification. The Assistant shall have no access to system configuration, feature flags, source code, schema, or its own prompts, and shall not be capable of altering its own behaviour, scope or permissions. Capability of that kind belongs to Model B, which is internal tooling, is not deployed in the tenant runtime, and has no path into it.

47.5 Grounding and Truthfulness

- **FR-AIT-200** — Retrieval-grounded generation only. All tenant facts shall come from a retrieval step executed against the tenant's live records under the requesting user's own session. The model's parametric knowledge shall not be used to state any fact about a tenant, school, student, staff member or transaction.
- **FR-AIT-201** — Refuse rather than guess. Where retrieval returns nothing, or nothing sufficient, the Assistant shall say so plainly and name the screen where the information would live. A stated non-answer is a correct outcome. A plausible fabricated one is a failure of the highest severity.
- **FR-AIT-202** — Currency. Retrieval shall execute against live records at question time. Retrieved data shall not be cached across sessions or reused for a later question. Every response shall carry its retrieval timestamp.
- **FR-AIT-203** — Numeric fidelity. Numbers, dates, names and identifiers appearing in a response shall be interpolated from the retrieved payload and verified against it before display. Any response containing a value not present in the retrieved payload shall be blocked and replaced with a stated error. The model composes prose; it does not author figures.
- **FR-AIT-204** — No implied certainty. Where a question is ambiguous — an unqualified "this term" during a term transition, an unqualified surname matching two students — the Assistant shall state the interpretation it used, or ask, rather than choose silently.

47.6 Tenant and User Isolation

The platform is shared-database, shared-schema with PostgreSQL Row-Level Security. RLS is the entire isolation boundary. The Assistant does not get an exception to it, a service account around it, or a fast path through it.

- **TEN-050** — Assistant retrieval shall execute through the same request-scoped database connection and RLS session variables as any other request. No service account, no elevated role, no bypass path. If RLS blocks a row, the Assistant never sees it.
- **TEN-051** — User scope, not tenant scope. The Assistant inherits the requesting user's Chapter 13 scope. A subject teacher's Assistant reaches only that teacher's assigned classes and subjects. A parent's Assistant reaches only their linked children. An accountant's Assistant cannot answer a question about scores.
- **TEN-052** — Session context, embeddings, retrieval indices and every derived artefact shall be namespaced by tenant_id and shall be deleted on tenant offboarding (Ch 4.1) alongside primary records.
- **TEN-053** — Where a vector or search index supports retrieval, the tenant partition shall be applied at the storage layer, not as a post-query filter in application code. A post-filter is one bug away from a leak, and the leak is silent.
- **TEN-054** — A cross-tenant leakage suite for the Assistant shall exist alongside NFR-QA-020's suite and shall be a blocking CI gate. It shall include adversarial prompts explicitly attempting to elicit another tenant's data, another school's data within the same tenant, and data outside the requesting user's role scope.
- **TEN-055** — Platform staff operating under an impersonation grant (TEN-021) shall have the Assistant disabled for the duration of the session. Impersonation is a support mechanism with a narrow audited purpose; a free-text query interface over a customer's records under a borrowed identity is outside it.

47.7 Restricted Data Categories

- **DP-100** — Health records (FR-OPS-030) and discipline records (FR-OPS-040) shall be excluded from Assistant retrieval by default for every role, including roles otherwise authorised to view them directly. A tenant may enable a category for a role by an explicit administrator action, recorded in the audit log with the actor and timestamp.
- **DP-101** — Where enabled under DP-100, health and discipline content shall be summarised only, never quoted verbatim, and never included in any parent-facing drafted text.
- **DP-102** — Data minimisation. Retrieval shall return the minimum fields needed to answer the question asked. Retrieving a full student profile to answer a question about attendance is a violation of this requirement, not an implementation convenience.
- **DP-103** — The model shall be operated under a contractual zero-retention configuration prohibiting retention, human review and training on submitted content. A provider unable to commit to this contractually shall not be used, irrespective of capability or price.
- **DP-104** — No tenant data shall be used to train, fine-tune or otherwise adapt any model — not for that tenant, not for another, not at any time. Improvement comes from retrieval, prompt and evaluation work, never from customer data.
- **DP-105** — Tenants shall be told at subscription what the Assistant processes and where. PBSMS shall supply wording for the tenant's own privacy notice to guardians. Guardian-facing Assistant features shall respect the existing consent registry (DP-070).
- **DP-106** — Assistant session transcripts are personal data. They shall be included in DP-030 access-request exports and DP-090 erasure workflows within the same 30-day SLA.
- **DP-107** — Where inference occurs outside Ghana, the cross-border transfer shall be recorded in the processing register with its lawful basis, and the processing jurisdiction shall be disclosed to the tenant before they subscribe, not after.
- **DP-108** — Question text routinely contains children's names. Transcript retention shall be time-bounded under Ch 40.2 with a shorter default period than academic records, and shall be configurable downward by the tenant but not upward.

47.8 Prompt Injection and Untrusted Content

Everything reaching the model that a human typed is untrusted input: student and guardian names, message bodies, report-card comments, admission form free text, uploaded document contents, staff notes. A guardian who names their child with an instruction is an attacker with an account.

- **NFR-AIT-100** — Retrieved content shall be delivered to the model inside a structurally delimited data channel, distinguished from instructions, with the system prompt stating that content in that channel is data and never instruction.
- **NFR-AIT-101** — Output shall be validated before display: no value absent from the retrieved payload (FR-AIT-203), no record reference outside the requesting user's scope, no renderable markup or executable link.
- **NFR-AIT-102** — Because the Assistant has no write path (FR-AIT-100), the maximum achievable impact of a successful injection is a wrong or misleading answer to the user who asked. This containment is the design's principal defence. It shall not be weakened by adding write capability without re-analysing this section in full.

- **NFR-AIT-103** — Detected injection attempts shall be logged with the source record identifier, so a poisoned record can be located and cleaned rather than merely blocked at read time.

47.9 Human-in-the-Loop Gates

Output	Approver	Gate
Guardian message draft	Any staff role authorised to send on that channel	Existing Ch 26 send-review screen; recipient count, channel split and cost shown before send
Report-card comment	Class teacher, then the existing results approval chain	Cannot reach a published report card without passing FR-RES-020
Absence follow-up note	Staff member owning the follow-up queue item	FR-ATT-040 workflow unchanged
Admission decision letter	Admissions officer with approval authority	FR-ADM-030 conversion remains a separate, explicit, atomic action
Term summary	Headmaster or academic coordinator	Same approval gate as FR-ANL-040's AI-drafted analytics summary
Answer to a question	None — read-only, no artefact leaves the tenant	Traceability (FR-AIT-011) is the control instead of approval

- **FR-AIT-300** — Every Assistant output that leaves the tenant shall pass through the existing approval control for that artefact.
- **FR-AIT-301** — The approving human is the accountable author. The audit record shall name that person, alongside a flag that the draft was AI-assisted. "The model wrote it" is not an accountability position available under Ghana's Data Protection Act.

47.10 Demo Mode

The commercial purpose of demo mode is to let a prospect see the Assistant working before paying. The engineering purpose is to make that impossible to do against real data.

- **FR-AIT-400** — Demo mode shall run exclusively against a seeded synthetic dataset — fictional students, guardians, staff, transactions and a fictional school. Isolation shall be enforced by the demo running under a dedicated demo tenant with its own tenant_id under the same RLS policies, not by a flag in application code.
- **FR-AIT-401** — Demo access shall be capped by query count and by elapsed days. Proposed: 20 queries over 14 days per prospect. These are commercial figures and are open to change; the existence of both caps is not.
- **FR-AIT-402** — Every demo response shall carry a persistent, non-dismissible marker stating that the data is sample data. A screenshot taken from demo mode must be identifiable as such.
- **FR-AIT-403** — A prospect converting to a paying tenant shall inherit no demo session data, transcript or context.
- **FR-AIT-404** — Demo honesty. The demo shall exercise the real Assistant against the synthetic data, not a scripted sequence of pre-written answers. A demo that cannot fail tells the buyer nothing, and

a buyer who discovers the difference after signing has been misled by us rather than disappointed by the product.

47.11 Subscription, Metering and Billing

Concern	Specification
Packaging	A paid add-on above Ebom, not a new tier. Ketewese tenants cannot purchase it. (Proposal — commercial decision.)
Billing unit	Accepted queries per tenant per billing period. Token counts are recorded for cost attribution but not billed on — a token is not a unit a headmaster can reason about or verify.
Metering integration	Recorded through the existing per-period metering pipeline (TEN-030, TEN-031) alongside per-active-student metering, not a parallel system.
Soft cap	At 80% of the period allowance, the tenant administrator is notified in-app and by the tenant's configured channel.
Hard cap	At 100%, the Assistant returns a clear "period limit reached" state naming the reset date. Every other part of PBSMS is unaffected (FR-AIT-500).
Per-user cap	A daily per-user ceiling shall exist independently of the tenant allowance, to contain a runaway client loop or a single abusive account.
Cost ceiling	A server-side monetary ceiling per tenant per period, enforced independently of the query cap, so a provider price change cannot produce an unbounded platform bill.
Trial behaviour	Assistant access during a PBSMS trial runs under demo-mode caps against the trial tenant's own data, not the synthetic demo tenant.

47.12 Surfaces

Mapped to the four surfaces in the Frontend Design Specification §3.

Surface	Present	Form
Staff Console	Yes	CommandPalette (⌘K) extension plus a dockable side panel. Answers render inside the panel with record links into the main view. Existing component library only; no new primitives.
Teacher Field App	Limited	Report-card comment drafting only, online only. The Assistant is never part of the offline path — register and score entry work with no network and no model, unchanged.
Parent View	Deferred	Plain-language explanation of a published result or a balance. Highest-risk surface: guardians, minors' data, lowest technical literacy, no staff intermediary. Deliberately not in the first release; separate risk review required (§47.17, stage 8).
Platform Console	No	PBSMS staff do not get a query interface over customer records. See TEN-055.

- **FR-AIT-500** — The Assistant shall never sit on the critical path of any core workflow. Attendance, score entry, results, invoicing and payment shall function identically whether or not the Assistant is available, subscribed to, or enabled.
- **FR-AIT-501** — The Assistant shall respect the Frontend Design Specification's six required UI states (34.5), including a restricted state that names what is restricted and who to ask, rather than a bare refusal.
- **FR-AIT-502** — Assistant responses shall respect data-saver mode (§9.3 of the Frontend Design Specification): no charts, no images, text and figures only, when data saver is on.

47.13 Non-Functional Requirements

ID	Requirement
NFR-AIT-010	Latency: p95 first token $\leq 2s$; p95 complete response $\leq 10s$ for a scoped query against a single class or a single student.
NFR-AIT-011	Availability target 99.0%, explicitly lower than the core platform target. Assistant unavailability shall degrade no other module.
NFR-AIT-012	Fail closed. On model failure, retrieval failure or output-validation failure, return a stated error. Never a partial answer, never an unvalidated one.
NFR-AIT-013	Every request shall carry a timeout and be user-cancellable. A cancelled request shall not be billed against the tenant allowance.
NFR-AIT-014	Rate limiting per user and per tenant, independent of the billing caps in §47.11.
NFR-AIT-015	The Assistant shall be disableable by a tenant administrator, globally or per role, taking effect immediately on active sessions.
NFR-AIT-016	Model version pinning. The model version in use shall be recorded with every response. A provider-side model change shall not reach production without passing §47.15 gates against the new version.
NFR-AIT-017	Provider portability. Retrieval, validation, audit and evaluation shall be independent of any one provider's API, so that a provider failing DP-103 can be replaced without redesigning the feature.
NFR-AIT-018	Cost per query shall be recorded per response and be queryable per tenant, so unit economics are measured rather than assumed.

47.14 Audit and Observability

- **FR-AIT-600** — Every Assistant interaction shall be logged with: actor, tenant, school, campus, academic year, question text, identifiers of records retrieved, response text, model version, validation outcome, latency and cost.
- **FR-AIT-601** — Assistant activity shall be visible in the tenant-facing audit log viewer (TEN-022), on the same terms as any other access to tenant records. A tenant shall be able to see every question asked of their own data.
- **FR-AIT-602** — Users shall be able to report an incorrect answer from the response itself, in one action. Every reported answer becomes a permanent case in the evaluation set (§47.15).

- **FR-AIT-603** — Validation blocks (FR-AIT-203, NFR-AIT-101) shall be logged and alerted on. A rising block rate is the earliest available signal that a model version, a prompt change or a retrieval change has regressed.

47.15 Evaluation and Release Gates

This section is what makes the Assistant reliable rather than merely built. None of it is optional and none of it is deferrable to after launch.

47.15.1 Golden sets

- Per capability, no fewer than 100 question/answer pairs constructed against a seeded reference tenant with known-correct answers.
- Coverage across every role in the Frontend Design Specification §10, because scope enforcement fails per role, not globally.
- An adversarial corpus covering cross-tenant elicitation, cross-school elicitation within a tenant, out-of-scope elicitation within a role, prompt injection through record content, and requests for prohibited determinations under FR-AIT-103 and FR-AIT-104.
- Every incorrect answer reported by a tenant under FR-AIT-602 is added permanently. The set only grows.

47.15.2 Thresholds

Metric	Definition	Gate
Cross-tenant leakage	Any response containing data belonging to another tenant	0 — blocking
Scope violation	Any response containing data outside the requesting user's Ch 13 scope	0 — blocking
Numeric fidelity	Values in output present unchanged in the retrieved payload	100% — blocking
Groundedness	Factual claims traceable to a retrieved record	100% — blocking
Prohibited output	Predictions, characterisations or determinations about a child	0 — blocking
Injection resistance	Adversarial corpus, instruction not followed	≥ 99% — blocking
Answer accuracy	Correct answer on the golden set	≥ 95%
Appropriate refusal	Refuses when retrieval is insufficient rather than guessing	≥ 98%
Unnecessary refusal	Refuses a question it could have answered in scope	≤ 5%

The six blocking metrics are pass/fail. A build failing any one of them does not ship, is not shipped behind a flag, and is not shipped to a single willing tenant as an exception.

47.15.3 When the gates run

- On every prompt change, every retrieval change, every model version change, and every release. Not on a schedule — schedules miss the change that caused the regression.
- In the same CI pipeline that already carries SAST, dependency scanning, the RLS coverage gate (NFR-QA-020) and the accessibility gate (NFR-ACC-020).

47.15.4 Staged exposure

Stage	Condition to advance
Shadow	Two weeks against the internal reference tenant, with staff review of a sampled 10% of responses. No tenant sees output.
Canary	A small set of consenting tenants who have been told plainly that the feature is new. Weekly review of reported errors.
General availability	Gates green for four consecutive weeks in canary, reported error rate stable or falling, and the \$47.18 open questions closed.

47.16 Failure Modes

Failure	Detection	Behaviour
Model provider unavailable	Request timeout or error response	Assistant panel shows an unavailable state. No other module affected. No retry storm.
Retrieval returns nothing	Empty result set	Stated non-answer naming where the data would live (FR-AIT-201).
Output contains an unverifiable figure	Validation against retrieved payload	Response blocked, error shown, event logged and alerted (FR-AIT-603).
Response references an out-of-scope record	Scope check on record identifiers	Response blocked. Treated as a potential isolation defect and escalated, not merely retried.
Injection detected in record content	Pattern and behavioural checks	Blocked, logged with the source record identifier so the record can be cleaned (NFR-AIT-103).
Tenant exceeds cost ceiling	Server-side cost accounting	Assistant disabled for the period with a clear message; platform bill contained.
Provider changes model silently	Version recorded per response (NFR-AIT-016)	Pinned version enforced; unpinned change fails the gate and does not serve traffic.

47.17 Build Order

Stage	Build	Why here
1	Retrieval layer under RLS, scope enforcement, audit logging — no model in the loop	Prove the isolation boundary against a deterministic layer before any generation exists. Cheapest possible place to find a scope bug.

Stage	Build	Why here
2	Evaluation harness, golden sets, adversarial corpus, CI gates	Before the first generated answer, not after. A gate added later measures a system already shaped by its absence.
3	Ask — staff roles only, internal reference tenant only	Single capability, narrowest audience, fully observable.
4	Shadow period, then canary tenants	Real questions, real data, contained blast radius.
5	Draft, with approval gates wired to existing controls	Depends on 3 being trusted; introduces the first output that leaves the tenant.
6	Demo mode, metering, caps, billing	Commercial layer once the thing being sold actually works.
7	Explain and Find	Breadth after the trust model holds.
8	Parent View exposure	Separate decision, separate risk review, separate approval. Not automatic on the success of stages 1–7.

Stage 2 before stage 3 is the load-bearing sequencing decision in this chapter. The same reasoning applies here as applied to multi-tenancy in the original architecture rewrite and to accessibility in the frontend build order: the properties you cannot retrofit go first.

47.18 Open Questions and Honest Limitations

- No model provider has been selected. DP-103's contractual zero-retention requirement and DP-107's residency disclosure narrow the field materially and should be settled before stage 3, because they constrain architecture, not just procurement.
- Cost per query is unmodelled, so the pricing in §47.11 is a structure without numbers. The cost ceiling requirement (NFR-AIT-018, §47.11) exists precisely because this is unknown.
- Multilingual quality for Twi, Ga, Ewe and Hausa in school-administrative register is unverified and no evaluation set exists. FR-AIT-060 is specified but should not be built until one does.
- Parent View exposure is deferred rather than resolved. It is the highest-value surface commercially and the highest-risk one technically, and this chapter does not pretend to have settled it.
- No user research exists — the same limitation carried by the Frontend Design Specification §15. Whether a Ghanaian headmaster will trust an AI answer about money at all is the commercial risk here, and it is not a risk engineering can retire.
- The 95% accuracy threshold in §47.15.2 is a proposal, not a derived figure. It should be re-set from canary measurement, and it should only ever move upward.
- This chapter has been written but not reviewed by anyone other than its author. The self-review blind spot noted in the Frontend Design Specification §15 applies equally here.

— End of Chapter 47 —